Use of phase-noisy laser fields in the storage of optical pulse shapes in inhomogeneously broadened absorbers

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It has been demonstrated [see, e.g., W.R. Babbitt and T. W. Mossberg, Opt. Commun. **65**, 185 (1988)] that coherent optical processes can be employed to store and reproduce temporal sequences of optical data, thereby providing a mechanism for advanced optical memories. We find that excitation pulse phase noise can be used to extend the range of experimental conditions under which the storage process is effective and discuss the use of phase noise to achieve secure data storage.

Optical memories have been proposed[1,2] in which information is addressed on the basis of both spatial and spectral degrees of freedom. Such memories provide a unique means of achieving ultrahigh-density, near-molecular-scale data storage and have attracted considerable interest despite the fact that their implementation will likely require the cooling of storage materials to near-liquid-helium temperatures. Of the various types of frequency-selective optical memory (FOM) that have been suggested, only time-domain[2] FOM offers the potential of achieving ultrahigh input/output speed and ultrahigh storage density simultaneously. In a time-domain FOM, data are encoded on the temporal envelope of an optical pulse (the data pulse), and this pulse together with a write pulse is used to illuminate one spatial location of the storage material. As a consequence the envelope of the data pulse is stored as a frequency-dependent modification of the storage material's absorption profile. To recall data stored at a particular spatial location, the location is illuminated by a read pulse. The read pulse stimulates the emission of a signal pulse that has the same temporal envelope as the stored data pulse, provided that the read pulse is essentially identical to the write pulse. Time-domain FOM's employing brief or, alternatively, frequency-chirped write and read pulses have been demonstrated.[3-6] We explore here the use of phase-noisy write and read pulses in time-domain FOM's. Such pulses may be useful in lowering pulse-power requirements and in the design of encrypted memories.

The constraints on write and read pulses in time-domain FOM can be appreciated by considering an expression for the output pulse envelope $E_{\rm sig}(t)$. We find that[3]

$$E_{\rm sig}(t) \propto \int_{-\infty}^{+\infty} \chi_{\rm wr}(\tau) E_d(t-\tau)\mathrm{d}\tau, \tag{1}$$

where $\chi_{\rm wr}(\tau)$, the cross correlation of the read- and write-pulse envelopes, is given by

$$\chi_{\rm wr}(\tau) = \int_{-\infty}^{+\infty} E_r(t')E_w(t'+\tau)\mathrm{d}t', \tag{2}$$

and $E_r(t)$, $E_w(t)$, and $E_d(t)$ are the electric-field envelopes of the read, write, and data pulses, respectively. Expression (1) for the output pulse envelope is valid provided that the write and data pulses both occur within a time interval shorter than the optical dephasing time of the optical transition excited, that the areas of all the input pulses are less than $\sim\pi/2$, and that the inhomogeneous absorption bandwidth of the storage material is greater than the bandwidth of the data pulse.

Expression (1) states that the envelope of the output signal $E_{\rm sig}(t)$ is proportional to the convolution of the data-pulse envelope with $\chi_{\rm wr}(\tau)$. We can immediately deduce that $E_{\rm sig}(t)$ will be essentially identical to $E_d(t)$ when $\chi_{\rm wr}(\tau)$ consists primarily of a single sharp peak whose temporal width is shorter than the shortest feature of interest in the data pulse.

In this Letter we provide an experimental demonstration of the fact that identical, temporally long pulses that have a pseudorandom phase modulation are effective as read and write pulses. Since the read and write pulses must be identical in order to have a sharply peaked cross correlation, it is impossible to access information that is stored by using a phase-modulated write pulse without having detailed knowledge of the phase modulation employed.

In order to be effective in data-pulse storage and retrieval, phase-modulated write and read pulses must satisfy a number of constraints that follow from expression (1). We analyze these constraints for the case when the identical write and read pulses are rectangular with a real electric field of magnitude E_0 and a total duration of T. During each pulse the phase of the field is assumed to be switched randomly between 0 and π at times separated by a constant interval $T_b < T$. The number of possible instances at which the phase could be flipped during the duration of the pulse is given by $M \equiv T/T_b$. The cross correlation of a pair of identical phase-noisy pulses used in our experiment is shown in Fig. 1. One finds that $\chi_{\rm wr}(\tau)$ consists of a sharp peak with an approximate width of T_b and a broad pedestal. It follows that for high-fidelity pulse envelope reproduction, T_b must be made shorter than

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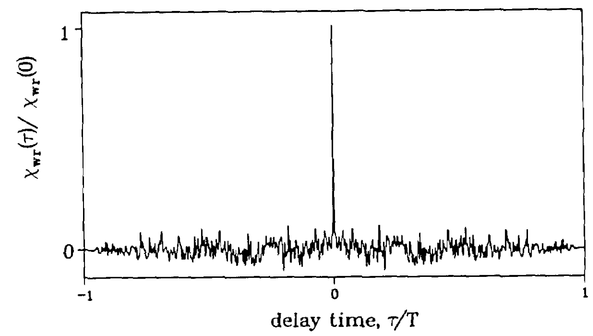

Fig. 1. Normalized electric-field cross correlation, $\chi_{wr}(\tau)$, of the identical phase-modulated write and read pulses employed in the present experiments. The phase is switched 300 times in a statistically independent manner between the values of zero and π. The width of the peak is approximately equal to the interval between phase shifts, which in this case is $T/300$, where T is the duration of the pulse envelope. The cross correlation was numerically evaluated from the measured-phase modulation pattern.

the shortest temporal feature of interest in the data pulse. Note in Fig. 1 that $\chi_{wr}(\tau)$ is nonzero for $|\tau| \leq T$. The broad pedestal in $\chi_{wr}(\tau)$ produces a temporally broad background in the output signal.

We estimate the size of the background relative to the recalled data pulse for a simple case in which the data pulse consists of a single square pulse (one bit of information). Let the amplitude and duration of the data pulse be denoted A_0 and T_d, respectively. First, using standard statistical arguments, one finds that

$$\chi_{wr}^{0} \equiv \chi_{wr}(0) \simeq E_0{}^2 T_b M \tag{3}$$

and (for $T_b < |\tau| < T$) that the root-mean-square value of χ_{wr} is given by

$$\chi_{wr}^{rms}(\tau) = E_0{}^2 T_b \sqrt{M'}, \tag{4}$$

where $M' = (T - |\tau|)/T_b$. For $|\tau| > T$, χ_{wr} is equal to zero. Note that when the read and write pulses consist of different random noise sequences, the mean value of the cross-correlation function is equal to zero for all τ.

In qualitatively evaluating the convolution of expression (1), we represent $\chi_{wr}(\tau)$, for simplicity, by (i) a rectangular peak of height χ_{wr}^{0} and width T_b centered at $\tau = 0$ and (ii) a noisy background in the range $-\infty < \tau < \infty$ that switches randomly between $\pm E_0{}^2 T_b M^{1/2}$ every T_b. The pedestal assumed in this simplified model is not identical to the real one described by Eq. (4); however, the assumed pedestal leads to essentially the same results as the real pedestal.

The magnitude of $E_{sig}(t)$ corresponding to the reproduced data pulse (the data signal) can be predicted from expression (1) using the data pulse shape and part (i) of the assumed correlation function above. One finds that the data signal has a magnitude

$$S \simeq A_0 E_0{}^2 T_b{}^2 M. \tag{5}$$

The portion of the output signal corresponding to the background noise can be estimated by evaluating the integral of expression (1) at values of t for which the peak of χ_{wr} and the data pulse do not overlap. In the present case the magnitude of the signal background is approximated by inserting part (ii) of the assumed correlation function above and the data pulse envelope into expression (1) and evaluating the result. Assuming that $T_d \gg T_b$, one finds that

$$B \simeq A_0 E_0{}^2 T_b{}^2 \left(\frac{M T_d}{T_b}\right)^{1/2}. \tag{6}$$

The ratio of the data-signal intensity to the background intensity is thus given by

$$\left(\frac{S}{B}\right)^2 \simeq \frac{T}{T_d}. \tag{7}$$

We conclude that good contrast between the data signal and the signal background is obtained by making the write and read pulses much longer than the duration of the data pulse, i.e., by making $T \gg T_d$. In the case of a multibit data pulse, it can be shown that the signal-to-noise ratio can still be determined from relation (7), provided that T_d is interpreted as the total length of time that the data bits are high. The assumption that $T_d \gg T_b$ made prior to relation (6) was made only for convenience in applying statistical arguments. Relation (7) should remain essentially correct even when T_d and T_b are of comparable magnitudes.

Our experimental apparatus is shown schematically in Fig. 2. The pulse-shape information was stored in a 2 at. % Eu^{3+}:Y_2O_3 crystal on the $^7F_0$–$^5D_0$ transition of Eu^{3+} (transition wavelength 580.8 nm) at liquid-helium temperature ($\simeq$4.5 K). At $\simeq$4.5 K the homogeneous relaxation time on this transition has been measured to be $\simeq$130 μsec.[7] The storage time in the material, determined by the thermalization time of ground-state sublevels, is of the order of hours.[4] Periodically the sample temperature was raised to approximately 20 K for 1 min to thermalize the ground-state sublevel populations. The write, data, and read pulses were acousto-optically gated from the output of a single-mode cw dye laser, and their respective pulse powers were 2.4, 1.4, and 2.4 mW. The identical write and read pulses (duration $\simeq$10 μsec) were phase modulated by an acousto-optical modulator (AOM2) that was driven by a random-noise-modulated rf signal. The rf signal was generated by mixing a 500-MHz carrier frequency with the output of a shift-register noise source, which produced phase shifts at a 30-MHz rate. By using feedback techniques, the shift register was configured to generate a repetitive sequence of 1024 pseudorandom bits. A portion (300 bits long) of

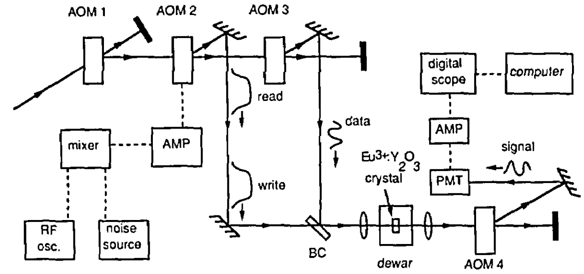

Fig. 2. Experimental setup. BC, beam combiner; PMT, photomultiplier tube; AMP, amplifier; RF Osc., rf oscillator.

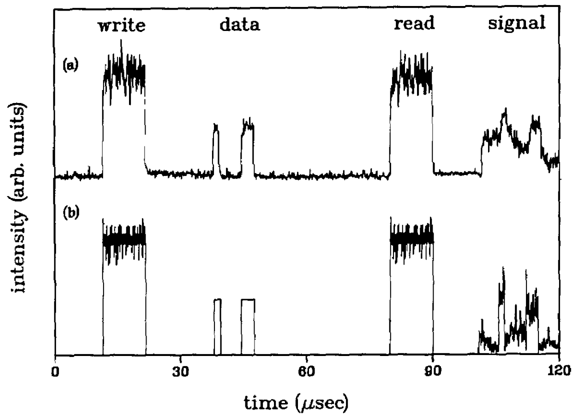

Fig. 3. Demonstration of faithful pulse-shape reproduction using write and read pulses with identical pseudorandom phase modulation. (a) Measured intensity profiles of the write, read, and data pulses along with the output signal produced. (b) Simulation of the experimental results using the measured pulse parameters and expression (1).

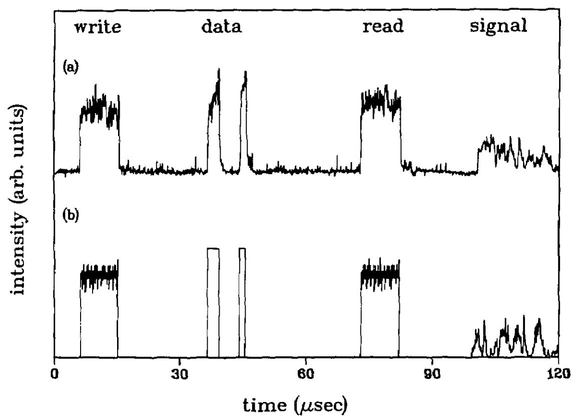

Fig. 4. (a) Output signal generated in the case of write and read pulses with nonidentical pseudorandom phase modulation. (b) Simulations of the experimental results using the measured pulse parameters and expression (1).

this sequence was sent to the mixer. The data pulse, which consisted of two subpulses (of $\simeq$1.5- and $\simeq$3-μsec duration and separated by $\simeq$10 μsec), was created by acousto-optic modulator AOM3. The pulsed laser beams were rendered collinear and were focused in the center of the Eu^{3+}:Y_2O_3 crystal to a spot size of $\simeq$70 μm. Acousto-optic modulator AOM4 was used to deflect only the signal pulse onto a photomultiplier detector. From relation (7) we expect that the signal-to-background ratio will be $\simeq$2 for our experiment, in which $T = 10$ μsec and $T_d = 4.5$ μsec.

In Fig. 3 we present the results of experimental and theoretical studies of data-pulse storage and reproduction by using identical phase-modulated write and read pulses; the curves represent intensity profiles. In Fig. 3(a), which represents experimental measurements, the signal was averaged over 250 events. It is clear that a replica of the data pulse appears in the output signal and that the signal-to-background ratio agrees approximately with the estimated value given above. In Fig. 3(b) we show the output pulse calculated according to expression (1) using measured excitation pulse parameters. The qualitative agreement with experiment is clear. The scales for the input pulses and the output signal are not the same in Figs. 3(a) and 3(b). The output signal was measured to be approximately 0.03% as intense as the data pulse.

We performed a second experiment to demonstrate that the pulse-shape information is not faithfully retrieved when different phase-noisy sequences are imposed on the write and read pulses. This was accomplished experimentally by selecting different pseudorandom segments from the output of the shift register. In this case the cross correlation of the write and read field amplitudes is not sharply peaked, and hence we expect that the data will not be recalled. The experimental results are shown in Fig. 4(a), where it is seen that the output signal does not contain a reproduction of the data pulse. A simulation for these experimental conditions is shown in Fig. 4(b).

In conclusion, we have successfully demonstrated that temporally long, phase-noise write and read pulses can be used to perform storage and retrieval of optical pulse-shape information. Temporally long phase-noise pulses are useful in that they (like frequency-chirped pulses) require less power than the brief Fourier-transform-limited pulses that are frequently employed in studies of storage and retrieval and because they can in principle form the basis of an encrypted memory. In addition, the results presented here imply that one can store a multitude of data pulses at a single spatial storage location simply by employing different phase-modulation patterns in the corresponding write pulses. A particular data pulse can be recalled by using a read pulse that is identical to the corresponding write pulse.

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